

Constrained Decoding for Robust Text-to-SQL: Pre-Proposal

Haiyun Lyu, Hao-Tien Cheng, Jeongsik Park, Manu Goel,
Wanting Jiang, Zihan Xie, Zhuoning Wu

University of Southern California

{haiyunly, haotien, jeongsik, manugoel, wantingj, xiezihan, zhuoning}@usc.edu

1 Motivation

Many real-world data pipelines require strictly valid SQL queries, yet unconstrained language models frequently generate syntactically invalid results. Such structural errors limit the reliability of language models in code and query generation, even when semantic intent is correct. Constrained decoding enforces grammar- and schema-level validity during inference, directly preventing invalid outputs. As an inference-time method, constrained decoding provides a lightweight and modular alternative to architectural changes or retraining.

2 Methods

2.1 Dataset

We use WikiSQL¹ and Spider 2.0-Lite² to evaluate NL2SQL performance across different levels of real-world complexity. WikiSQL enables controlled evaluation of semantic parsing correctness under simplified single-table queries, while Spider 2.0-Lite reflects enterprise-style database environments with multi-table schemas and realistic SQL dialects, enabling robustness evaluation under practical complexity. Given computational constraints, we may randomly sample 10% of the WikiSQL training set.

2.2 Model and Training

We will fine-tune lightweight open-source language models selected from the Qwen and LLaMA families using quantization for efficient training, and compare them against off-the-shelf counterparts. Efficient inference will be supported by vLLM.

2.3 Constrained Decoding

We will study constrained-decoding frameworks that enforce hard structural validity during generation, including Outlines, XGrammar, and LM Format Enforcer, which restrict token generation to grammar-compliant sequences, with efficient inference supported by vLLM.

¹<https://github.com/salesforce/WikiSQL>

²<https://spider2-sql.github.io/>

Dataset	train/dev/test	Description
WikiSQL	56k / 8k / 16k	Single-table queries with simple SELECT-WHERE structure; large-scale supervised NL2SQL benchmark.
Spider	0 / 0 / 547	Multi-table schemas, complex predicates, and realistic SQL dialects reflecting enterprise-scale database environments.

Table 1: Datasets for this work. ‘Spider’ refers to the Spider 2.0-Lite.

2.4 Evaluation

Recent state-of-the-art system will be used as baselines for comparison. We will evaluate our off-the-shelf prompting, fine-tuned models, and constrained decoding methods against the baseline. Evaluation will primarily rely on execution accuracy (EA). Qualitative error analysis will further examine failure modes such as incorrect schema selection, aggregation errors, join reasoning failures, and semantically incorrect but syntactically valid SQL outputs.

3 Expected Results

(1) Fine-tuned models are expected to achieve higher execution accuracy than off-the-shelf models. (2) Constrained decoding is anticipated to improve syntactic validity and SQL executability, particularly in complex multi-table settings, and its combination with fine-tuning is expected to provide the strongest robustness and generalization in realistic database environments. (3) We expect constrained decoding to yield more stable generation with reduced sensitivity to stochastic decoding parameters, while introducing moderate inference-time overhead due to hard structural constraints, reflecting a correctness–efficiency trade-off.

4 Future Plans

Preliminary results will be obtained by the midterm milestone (Mar. 23), followed by complete experiments, figures, and analysis for the final presentation (Apr. 13-) and report (May. 02).